



Congratulations on the purchase of your T-37. We hope that she gives you years of pleasurable sailing. Included with this booklet is a warranty registration form. Please fill it out and return to Tartan Marine so you can be enrolled in the ranks of Tartan owners.

If questions arise about your boat which your dealer cannot answer, call Bill Seifert here at the office (216 354-5671. Comments about your dealer should be directed to me.

Happy Sailing!

Ben Garrett

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I. COMMISSIONING

BEFORE LAUNCHING:

Check that the engine and head have been de-winterized, and that all drain plugs are in place. The engine fresh water cooling system is filled with permanent anti-freeze solution at the factory. Check for evaporation and add more anti-freeze solution if required.

Inspect all thru hull connections, and open valves.

Mark propellor shaft with fixed prop vertical and folding prop with blades opening horizontally. For best sailing speed, set shaft to this position while sailing. Never leave engine out of gear and allow the shaft to rotate while sailing.

Install fully charged batteries.

Check engine and transmission oil level, and level of coolant in fresh water cooling system.

Note the location of the draft marks on the hull - refer to "Draft Marks" section.

Secure centerboard if needed - refer to centerboard instructions.

When picking up boat, note proper location for straps, and mark rail so that the straps can be easily positioned upon haul out.

AFTER LAUNCHING:

Immediately check bilge for water.

When the engine is initially started, check that water is coming out the exhaust. If water does not come out the exhaust, check that the engine water thru hull is open and not blocked, then check the water pump impellor.

After engine has run in gear for several hours, adjust the packing at the shaft stuffing box. Avoid over tightening - it should drip a little water.

After spar has been stepped, check engine alignment.

STEPPING THE MAST:

Because the centerboard pennant exits through the mast, a little advance planning is necessary before stepping the mast.

If there is not sufficient water depth (i.e., at least 8') at the spar stepping area, it will be necessary to

pass a line under the boat from the bow, working it aft until it is on the third stanchion from the bow. Draw the line up tight under the boat and secure to the stanchion bases port and starboard. This line will support the centerboard when its lifting pennant is uncleaned.

There is a section of PVC pipe shipped loose with the boat. Slide the centerboard pennant through the pipe and screw the coupling on the pipe into the thread on the top of the stainless pipe extending through the mast step. The PVC pipe will extend above the top of the partners by several inches.

Before the spar is stepped, install a messenger from the centerboard exit box on the starboard aft quarter of the mast down inside the spar to the butt.

Read "Adjustment of Mast Step", "Mast Boot" and "Mast-head Sheave" sections before stepping.

When the mast is hanging vertically over the boat, attach the spar messenger to the end of the centerboard pennant carefully so it will pass through the spar exit box. As the spar is lowered into the boat, keep a strain on the messenger so it does not get wedged inside the spar. Before the spar is fully stepped, the centerboard pennant should exit from the spar exit box.

If the centerboard pennant should be lost in the stepping operation, it can usually be fished out by removing the exit box in the spar and using a hooked piece of stiff wire.

PRELIMINARY TUNING:

As soon as the spar is stepped in the boat, and the headstay, backstay and upper shrouds are loosely attached to their respective chainplates, the spar should be chocked in the partners. Refer to "Chocking the Main Mast". This job is considerably easier before the rigging is tensioned.

The next step should be to establish the masthead in the center of the boat. Use the main boom topping lift and a tape measure to measure to the upper shroud chainplates alternately on each side and adjusting the turnbuckles so the reading on the tape is the same on each side. Once the masthead is centered, taking up an equal number of turns on each upper shroud turnbuckle will tension the uppers and maintain athwartships position.

After tensioning uppers, check that the spreaders have not been pulled down from the upward angle (about 5 degrees) established by the spreader bases. If the spreaders do not have an upward angle, go aloft in a chair and move the outboard end of the spreader upward to the angle of the spreader bases.

A toggle is provided to go in between the stemhead and link plates at the bow. Generally, the headstay goes

in the second hole down in the plates, but this may be changed dependent on rake desired. It may be necessary to attach a genoa halyard to the jib tack shackle and tension the halyard with the winch to pull the spar forward to get the headstay in the required link plate hole. The masthead should not be pulled so far forward that tension on the backstay does not result in a straight mast.

It is absolutely impossible to make any judgment regarding mast rake when the boat is not on an absolutely level flotation plane. Check the section "Draft Marks" for the procedure of leveling the boat fore and aft. With the boat exactly floating level, a 3' or longer carpenter's level held against the aft side of the mast will give a good indication of rake, by the simple geometric method of similar triangles. For example, the distance from the gooseneck height to the masthead is approximately 43'. If the bottom of a 3' long level must be held out 1/4" from the spar when the level reads exactly vertical, then the equation becomes:

$$\frac{.0208''}{3} = \frac{X}{43'} \quad \text{therefore } X = \frac{(.0208)(43)}{3}$$

$$*1/4'' = .0208'' \quad X = .298' \text{ or about } 3-5/8'' \text{ of rake}$$

The fullness of the sails will be the determining factor on rake. We have found that a difference of 8" in rake is not uncommon to optimize a boat's performance to the cut of the sails. An aft rake of about 3" is average for a fairly full mainsail.

Adjust the forward and after lower shrouds so the mast is straight fore and aft. The lowers should be a little looser than the uppers.

FINAL TUNING:

With the boat sailing hard on the wind, sight up the main mast. If the masthead appears to be falling off to leeward with a bow to windward in the middle, then the lower shrouds are too tight. Conversely, if the middle of the mast is sagging to leeward, the lowers are too loose.

Always tack and adjust turnbuckles on the leeward side. Never adjust a windward side turnbuckle while sailing.

Any deviation from straight fore and aft can be taken up by adjustment of either the forward or aft lower shrouds.

Many tacks and mast sighting will be necessary to achieve perfect tune, as adjusting one side has an effect on the other side. Many owners do not try to achieve a perfectly straight mast until the rigging has had a chance to relieve itself of the initial stretch inherent in wire.

Generally a few sails on windy days will stretch the rigging so a final perfect tuning may be made.

Especially with new rigging, the prudent skipper will check the mast for straightness on every voyage. After sailing in especially heavy winds, allow the rigging some time to "normalize" before adjustment.

CHOCKING THE MAINMAST:

After the mast is in the boat, rig a line from the mast up to a snatch block held in the stem head. Lead the line either to a halyard winch or a cockpit winch and pull the mast as far ahead in the partners as possible. Insert the thick piece of rubber provided, centered in the aft portion of the space between the partners and the mast. Ease off on the line pulling mast ahead in order to hold the rubber in position. Now rig a line extending around the forward part of the mast to snatch blocks attached in the rail, and to the cockpit winches. Have two strong people grind both cockpit winches simultaneously in order to pull the mast aft and compress the rubber chock on the aft side of the mast. Spray a small quantity of silicone or other non-petroleum lubricant onto the mast, the partners, and the chocking material. Slide the chocking material into the forward space between the partners and the mast. It will be a tight fit. Release the line holding the spar aft.

Hose clamps are provided which can be hooked into each other to encircle the spar and rubber chocking material at a position just below the partners. These clamps must be installed and tightened to hold the chocks in place while sailing.

ADJUSTMENT OF MAST STEP:

The mast step is adjustable fore and aft. The bolt which goes through the mast step in the center of the spar should be snugged down, but not torqued before the mast is stepped. After the mast is tuned, the outer bolts which are accessible with the spar in place should be tightened down hard to prevent movement of the mast step. Position of the step with a 3" rake will be about 3/8" aft of center adjustment.

ADJUSTING CENTERBOARD TRAVEL:

After the spar is stepped, raise the centerboard pennant through the cheek block mounted on deck and through the round fairlead by the dodger coaming. Lead the pennant directly to the winch from the fairlead, and raise the board fully. You will hear the board hit the top of the trunk, indicating it is fully up.

With the board fully up, measure back on the centerboard pennant 27-1/2" from the aft side of the round fairlead. Mark the pennant at this point.

Jam off the pennant so the board will not drop. There is a single non-swiveling block shipped with the boat. Tie a bowline in the centerboard pennant to hold this block. The forward end of the bowline should end up at the mark made in the step above. Excess pennant line may be cut off.

Reave the centerboard tag line through the single block and tie a figure eight in the end of the tag line. The tag line should be used around the winch to raise the board, as it gives a 2:1 power ratio.

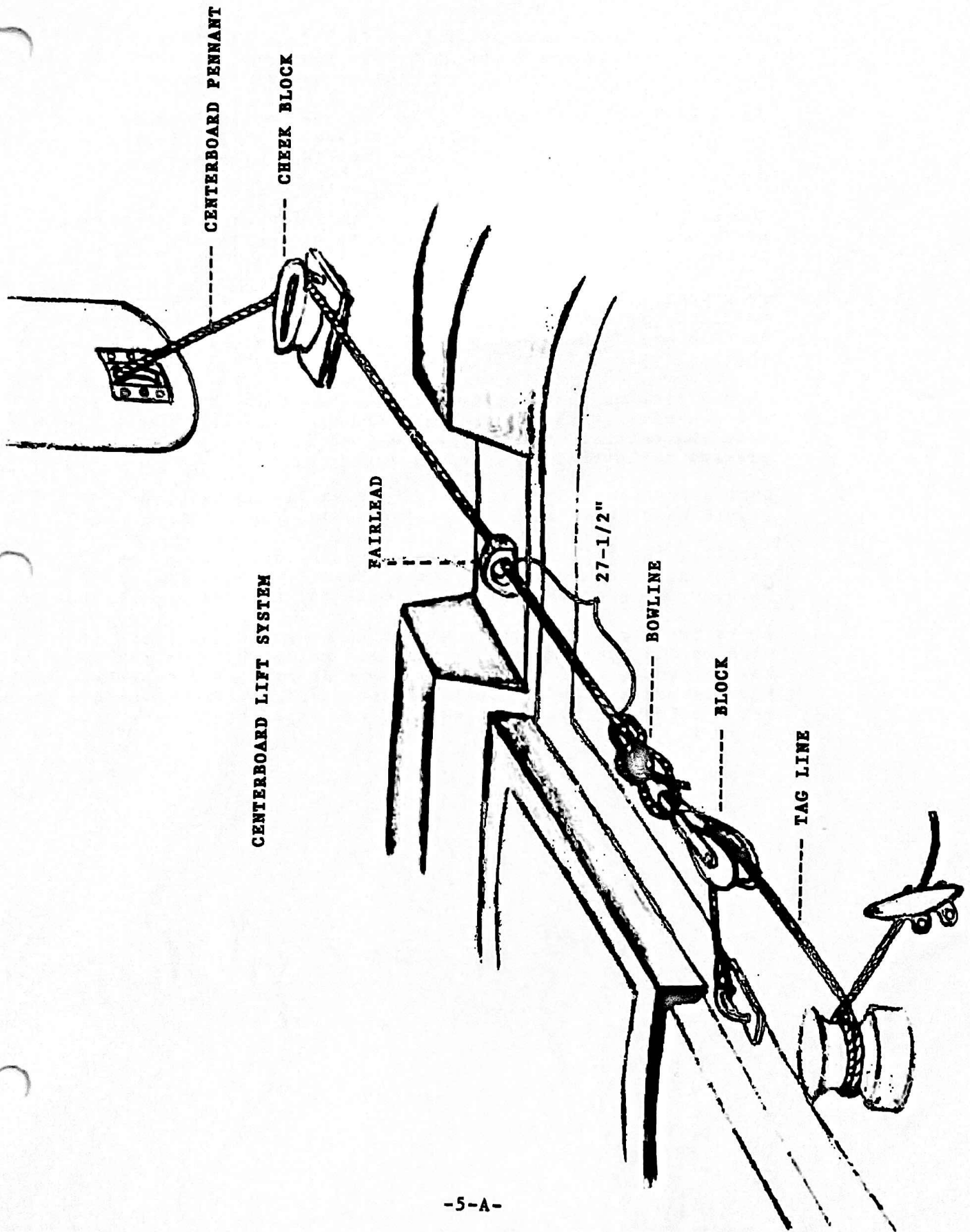
STEERING:

The T-37 is equipped with an Edson Radial Drive pedestal steering system. Information on maintenance of this system is included elsewhere in this manual.

Additionally, an emergency tiller is provided. The tiller is in two pieces. One-half of the emergency tiller fits inside the rudder post, and may be left in this position. The tiller handle is an aluminum tube, which can be stored in the lazarette and kept available for immediate use.

The emergency steering system does not allow as much rudder angle as the pedestal steering system, and the force on the tiller can be considerable, as the mechanical advantage in the pedestal system is not used. Steering with the emergency tiller may be more convenient if the steering wheel is removed from its hub. This is readily done by unscrewing the nut which holds the wheel on the hub, and pulling the wheel off the hub. Each time the steering system is inspected, check that the wheel has not seized on the hub, and can still be readily removed.

Be certain to try out the emergency steering system while you are becoming acquainted with your boat, and develop the procedures necessary to rig the system quickly in the event of failure of the pedestal system.



RIGGING TIPS:

Always lubricate turnbuckle threads before tightening them. Rod Stephens likes Anhydrous Lanolin. If that is not available, Lubriplate or similar grease works well.

In general, clevice pins should be inserted in toggles and turnbuckles so the cotter pin ends up either facing aft or inboard. This will result in less chance of snagging.

Cotter pins should only be opened up 10°, per Rod Stephens, so they can be pulled out readily in an emergency, and re-used.

In stepping the mast, it is generally easier if the toggles and link plates are attached onto the deck, and the turnbuckles are opened evenly while attached on the ends of the rigging.

When attaching the rigging onto the mast, insert the clevice pins so the cotter pins are in against the spar, open the cotters, then apply a dab of silicone sealer to prevent the cotter pins from snagging sails.

During commissioning, the length of the genoa halyard should be checked with the headsails for the boat.

Ideally, the sails should have adequate luff length, or extending pennants, so there are three to four turns of wire on the genoa halyard winch with the sail fully hoisted.

It is important that there always be at least two turns of wire on the genoa halyard winch, and sails which do not have adequate luff length to allow this condition must have head pennants added to them. Under no circumstances should genoa halyard tension be taken on the rope tail of the halyard.

II. GENERAL INFORMATION

MASTHEAD:

The sheaves in the masthead are made of aluminum. Before the spar is initially stepped, disassemble the sheaves and clean the sides of the sheaves and the masthead to remove shipping dirt. Grease the sides of the sheaves with a heavy grease to prevent their galling. This disassembly and greasing procedure should also be a part of Spring commissioning.

MAST BOOT:

A vinyl mast boot is provided, which must be installed around the mast by sliding it on from the bottom end of the spar before the spar is stepped. Once the spar has been properly chocked and the boot is smoothed down over the mast partners, a bead of hardening silicone or other bedding compound around the top of the mast boot and the spar will assure water tight integrity. Also, remember to fill the sail track with silicone bedding at the top of the boot.

DRAFT MARKS:

At the bow and stern there are impressions of screws in the gel coat surface. These are NOT imperfections, and are placed in the hull to give a reference point for establishing level flotation.

The boats float a little down by the bow. It will be necessary to put several people in the cockpit to level the boat, as would happen when sailing. Measure from the center of the screw head to the water both bow and stern, and adjust the cockpit personnel so that the measurement at the bow and stern are the same. The exact amount of the measurement is not important, just so the measurement is equal both places.

HATCHES:

Non-skid tape is a good idea on the translucent deck hatches as they become slippery when wet. The top of the hatch is made from plexiglass, which will be scratched by ground-in sand, coral, etc. If more privacy is desired, sand the inside of the hatch cover with #120 sandpaper which will "frost" the surface.

PORTLIGHTS:

The ports are glazed with plexiglass and Lexan, noted for its high impact resistance. Gritty cleaning agents, such as cleanser, will scratch. Use only mild soap and water to clean the ports. Some chemical solvents, notably Acetone, will also injure the port lights.

COMPASS:

The binnacle compass supplied with your T-37 does not have built-in compensators installed because built-in compensators can induce heeling errors.

If compensation of the compass becomes necessary because deviation causing metals cannot be moved away from the compass, correction should be made with bar magnets placed about the vessel. In the event that your compass cannot be corrected by removal of deviation causing metals, and bar magnets are not available, the "built-in" type of compensating magnets are provided with the vessel.

Compass correction is an owner obligation.

HEAD:

It is always good seamanship to close the intake and discharge seacock (if installed) for the head when leaving the vessel. Also be certain, while sailing, that the flush control valve or lever on the head is not left in the flush position to prevent the head from filling with water which will spill as the boat heels.

ELECTRICAL SYSTEM:

The T-37 has two totally independent electrical systems.

The 110V A.C. system is active only when the vessel is plugged into shore power via a shore power cord. The plug which fits the power inlet fitting installed on the cockpit coaming is included with the boat. Have a competent electrician wire this plug to your cord, and check the polarity of the dockside outlet. Never use an adapter to plug your shorepower cord into a two prong outlet.

Two circuit breakers will be found on the aft side of the bulkhead outboard of the navigation table. One circuit breaker controls the hot water heater, the other the outlets. A mis-wired cord or shore outlet can render these circuit breakers inoperative. Be sure to disconnect the shore power cord when working on the 110V wiring. Read and follow the instructions on the hot water heater in this manual.

The use of inexpensive automotive battery chargers which do not have a built-in isolation transformer can cause electrolysis to the vessel. Disconnect the NEGATIVE battery lead from the engine when using any charger other than a high quality marine charger with built-in isolation transformer.

A 12V D.C. electrical system is used throughout the vessel for lighting and operation of pumps and various

accessories. For any 12V current to be delivered, the following criteria must be met:

1. Charge in the battery
2. Master switch in Batt. 1 - Batt. 2 - or BOTH position.
3. Master circuit breaker on the electrical panel above the navigation panel - ON.
4. Circuit breaker for the individual appliance - (cabin lights, running lights, etc.) - ON.
5. Switch on the appliance (if there is one, such as cabin light on electric bilge pump) - ON.

The Battery Monitor installed on the electrical panel above the chart table will give an indication of the charge status of the battery to which the battery selector has been turned. The Battery Monitor will show a high reading when the engine is on and the alternator is charging.

The Ammeter located on the electrical panel will show the amperage being drawn from the batteries by the various electrical appliances.

The amount of charge going into the batteries is shown on the Ammeter in engine panel in cockpit. Generally, this meter will show a high rate of charge as soon as the engine starts, and the charge will taper off as the batteries come up to full charge. The voltage regulator automatically regulates the amount of charge going into the batteries, and reduces the level to prevent the batteries "boiling" over as they reach capacity. For this reason, even though the engine has a 55 amp alternator, charging the batteries for an hour will NOT put a full 55 amp hours back into the battery.

Batteries will last longer if they are kept charged during periods when they are not used. Be sure to check the water level in the batteries at least every two weeks. Adding water to a partially charged battery will lower the charge in the battery. Never add water to a battery which is charging, either via the engine alternator or a separate charger. Be careful in adding water so that the battery acid does not splash. Never add salt water to a battery. Most boat batteries have a shortened life from improper storage during lay up periods, lack of water, and the use of "quick" chargers.

FRESH WATER SYSTEM:

Fresh water is stored in one approx. 60 gallon tank located under the lower berth in the main cabin starboard side, and one 30 gallon (approx) tank located under the seat outboard of the table starboard side.

Both water tanks have inspection ports in the top of the tank. These may be opened to clean out the natural accumulation of scum inside a tank. Use caution that the ports are not excessively tightened. Newer boats have transparent inspection ports, and the water level should be observed while filling the tanks. The vents for the water tanks are below decks and the overflow when the tanks are full will run into the bilge. The overflow from filling tanks can be misconstrued to be indicative of a structural leak, since the overflow water also ends up in the bilge. Both tanks fill through a fill fitting located on the side deck. Each time a tank is filled, it is wise to wipe off the threads on the center plug and those on the deck fitting to remove dirt which may prevent a good seal. The starboard tank will fill first. In order to fill the port tank from the deck fill, both valves controlling from which tank the pressure pump takes its water must be OPEN. A more expedient method of filling the port tank is through the inspection port on the top of the tank.

When a tank has run dry, be sure to close the valve to the empty tank before opening the valve to the full tank. This will minimize the amount of air sucked in by the pressure water pump. When a tank has run dry, the pressure water system may have difficulty overcoming the air in the water lines. After the valves have been changed so that there is water available to the pressure pump, and the pump is turned on at the electrical panel, open a water faucet to allow air to escape. Eventually, there will be a trickle of water. When this happens, close the faucet momentarily to allow the pump to build up pressure, then open the faucet until a steady stream of water issues from the faucet. It may be necessary to repeat the process several times to bleed all the air from the system.

Read the CAUTION regarding the hot water system.

The water in your tanks may develop a taste after a period of time. This will happen to any water in any tanks, as it grows "flat". The addition of a commercial water preservation agent such as Sudbury Aqua Fresh crystals will greatly improve the taste of water stored for a long period.

HOT WATER:

Water is heated either by the engine heat exchanger or by an electric element when the boat is plugged into 110V A.C. power on shore.

*****CAUTION*****

When the water heater is operating from shore power, the pressure water system must be left on, and there must be water in the tank from which the withdrawal is connected. If a supply of water is not available to the water heater when the heater is on shore power, the element WILL BURN OUT.

When shore power is not available, operating the engine for 45 minutes to an hour will heat water sufficiently hot for a shower. This process happens automatically, provided the pressure water system is on and water is flowing to the water heater.

Before taking a shower, check that the sump pump is operational (refer to Sump Pump - in Plumbing section), or the shower drain water will overflow the sump tank and flood the bilge.

AUTOMATIC SUMP PUMP:

The shower and ice box drain into an independent sump tank located in the bilge. There is an automatic pump which drains this sump. In order for the automatic pump to drain the sump, the following must be activated:

Main Battery Switch - ON to battery 1, 2, or both.

Main Switch - on electrical panel - ON.

Sump Pump Switch - on electrical panel - ON.

Bilge Pump Manual/Sump Pump Automatic Switch - on panel - in Sump Pump Automatic position.

Water in sump tank - above about a 4" depth.

Electrical Bilge Pump Strainer and Hose - with strainer on bottom of sump tank.

With the above conditions met, the bilge pump should drain the water in the sump tank down to about 1", and then shut off. If the pump does not shut off, the probable causes are:

Cause:

Bilge pump manual/
sump pump automatic
switch in wrong position.

Bilge pump strainer
clogged.

Automatic control
switch line clogged.

Solution:

Turn to Sump Pump Automatic
position.

Remove from sump and clean
face of strainer.

Follow small tubing from air
bell in sump to the red topped
switch mechanism. Disconnect
tubing from back of switch
mechanism and blow hard into
tubing to clear tubing.

Cause (continued)

Automatic control out
of adjustment

Solution (continued)

Remove red cover on automatic bilge pump control unit and very carefully adjust screws which control contacts of micro switch so that the pump shuts off. This is a delicate control and some adjustments will be necessary until the "right" position is found.

The pump which drains the sump can also be used as a power bilge pump. Remove the hose and strainer from the sump tank, and place in bilge water. Move the "Bilge pump manual/Sump pump automatic" switch to the "Bilge pump manual" position. The automatic switch will be bypassed, and the switch on the panel labeled "Sump Pump" will turn the pump on and off.

TANKAGE:

The T-37 carries approx. 90 gallons of water, and approx. 50 gallons of fuel. The liquid in these tanks weights over one thousand pounds, and will affect the trim of the boat.

For short cruises, fill the starboard water tank only and do not fill the fuel tank all the way. Diesel fuel does loose some volatility and will grow bacteria, so it is better to not top the tank at the beginning of the season unless that fuel will be used quickly.

FUEL TANK:

Is located under the port quarterberth, outboard and behind the chart table. There is an inspection port in the berth top under the cushion which gives access to a shut-off valve. Since diesel engines require bleeding after they have been deprived of fuel, it is important the the shut-off valve be in the ON position (lever parallel to piping) anytime the engine is started.

The fuel tank utilizes an electric fuel gauge. Due to the shape of the tank, the readings of the gauge are not a direct reflection of the fuel in the tank. When the tank is half full, the gauge will read more than 1/2, for example. The prudent skipper will note the engine hours in the log each time fuel is taken aboard, and use the ship's log as an indicator of consumption.

When the fuel tank is nearly empty, motor sailing, especially heeled substantially on the starboard tack, may result in the fuel leaving the withdrawal tube momentarily, and the engine stopping.

In fueling, the prudent skipper will take the same precautions as with a gasoline engine, and close all ports and the companionway. Be sure to wash off all spilled diesel fuel, as it makes the decks very slippery. Clean the threads in the

fuel filler cap and deck plate each time fuel is taken aboard, and be sure the filler cap is screwed down tightly to prevent water entering the tank.

DIESEL ENGINE:

The most common cause of trouble with a diesel engine is contaminated or dirty fuel. Your boat is equipped with a primary fuel filter located on a bulkhead aft of the engine, and a secondary filter on the engine. Watch the primary filter for signs of dirt or water, and clean the filter accordingly. Refer to the engine manual for the changing procedure for the secondary filter on the engine. The wise skipper carries replacement filter cartridges.

Familiarize yourself with the bleeding procedures for the engine, and try bleeding it. The procedure only takes five minutes after you are acquainted with what to do, but can be exasperating to the uninitiated.

Diesel engines generate considerable heat. Always let the engine idle for a minute or two before shutting down. The blower may be used to rid the engine space of residual heat after the engine stops.

ENGINE OPERATION:

Because a diesel engine does not have an electrical ignition system, the oft troublesome key switch has been omitted. Anytime the master battery switch is on, the push button labeled "Start" on the engine instrument panel will cause the engine starter to turn over.

Before starting the engine, make certain that the stop knob (located below the engine instrument panel) is pushed in all the way. Also check that the gear shift (black handle on pedestal side) is in the neutral position (about the middle of its travel).

Advance the throttle a little, perhaps to 1/4, and push the "Start" button. In cold weather, the "Preheat" button may be used to help start the engine. Refer to engine manufacturers instructions.

The engine alarm system may sound for a few seconds after the engine has started. This system warns of low oil pressure and/or over-temperature. If the alarm does not go off soon after the engine has started, stop the engine and determine the source of the problem.

To stop the engine, reduce throttle to idle and pull out the engine stop knob located under the engine instrument panel. Be sure this knob is held out until the engine stops rotating, or it may re-start. After the engine has stopped, push the knob in again.

As with any engine, do not shift from forward to reverse at high engine RPM. Also, keep the engine in gear when sailing.

The engine owners manual contains a wealth of information about the engine. Take time to read the manual BEFORE you need the information due to a malfunction.

EXHAUST SYSTEM:

Your boat is equipped with a water lift principle muffler. This system cools the engine exhaust as it exits from the engine, reducing heat build-up, and substantially reduces exhaust noise.

If the flow of cooling water is interrupted, and the engine overheats severely, the rubber hose coming from the engine exhaust elbow may melt. Always check this hose after an occurrence of over heating.

There will always be a little water in the bottom of the water lift "pot". In Fall decommissioning, the pot should be drained using the drain plug, or anti-freeze should be added to the pot so that the residual water will not freeze.

If the engine should loose compression from sticking valves, bad piston rings, or other causes, and is cranked for a prolonged period, engine cooling water may build up in the pot. In this very unlikely situation, the cooling water intake should be shut off to prevent flooding the engine, or the drain on the pot opened.

Before engine cooling water is injected into the exhaust elbow, it runs up to a point above waterline where an anti-syphon valve admits air to the line when the engine is off, to prevent syphoning. Especially if operating in dirty water, the anti-syphon valve may clog, and fail to admit air at engine shut down. Check the operation of the anti-syphon valve frequently.

CHANGING THE CENTERBOARD PENNANT:

Material Required:

- 1 - new pennant - 3/8" diameter Dacron braided rope (preferably pre-stretched type) X 30 feet long.
- 1 - hose - heater hose will do, 1" inside diameter X 22 inches long.

The centerboard pennant is rope, and will eventually wear. The wear is most likely to occur on the underwater positions of the pennant, where inspection will require hauling or a swim under the boat.

It is NOT necessary to haul the boat to change a centerboard pennant if a swimmer can tie a bowline underwater. If a snake is not available, the exit box in the spar can be removed, and the end hardware removed from a lifeline so that the lifeline wire can be used as a snake. This may damage the lifeline, and an electrician's snake is preferable, about 20' long.

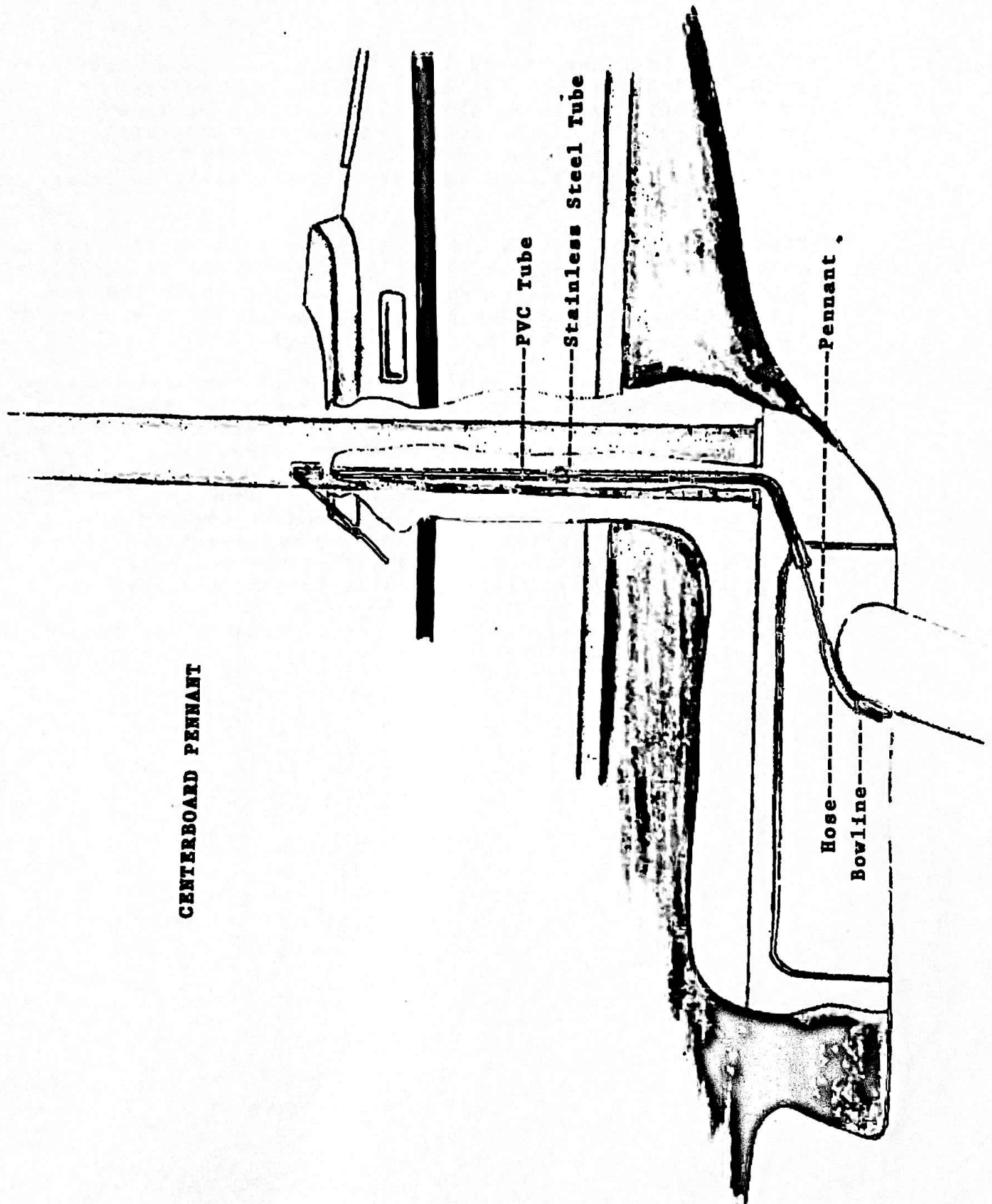
Feed the snake down the tube inside the mast until there is about 5' left on deck. Carefully tape the end of the new pennant onto the deck end of the snake. Overlap the two about 2 feet to allow for plenty of tape. Tie a stopper knot in the other end of the new pennant.

A swimmer can now grab the exposed end of the snake underwater which will be hanging out of the trunk, and bring this end back on board. The new pennant can now be pulled through and the tape removed from the snake.

The swimmer should now take the piece of hose and the end of the centerboard pennant which is coming up from the trunk underwater, slide the hose on the pennant and secure the end of the pennant onto the pad eye in the board with a bowline. The hose should be slid down to the bowline.

Adjust the length of the pennant on deck using the procedures in "Adjusting Centerboard Travel".

CENTERBOARD PENNANT



STOVE OPERATION:

Most apprehension about alcohol stoves comes from a lack of experience and knowledge of the basic working of the stove.

Alcohol stoves operate by vaporizing liquid alcohol to a gas. The gas is burned in much the same manner as in a home gas stove. Most problems are caused by improper preheating of the burner, which does not vaporize the alcohol.

A typical stove operating routine would be:

1. Fill the alcohol tank to about 2/3 full. A funnel with a hose attached makes this job considerably easier.
2. Close the filler cap securely.
3. Check that the stove shut-off valve (located on the bulkhead in front of the galley sink) is OFF.
4. Pump up the tank to about 10 lbs. air pressure. Some alcohol tanks will have a separate air pump, which operates like a bicycle pump.
5. Be certain that all stove burners are OFF.
6. Open the stove shut-off valve.
7. Open the burner control valve for the burner you wish to use, about one turn, and wait about three seconds. Watch the alcohol flowing into the pan under the burner, and do not overfill the little cup. THEN CLOSE THE BURNER VALVE.
8. Wait a few seconds, then ignite the alcohol in the priming pan under the burner. The resulting flame should not be more than 6" above the top of the burner. If there is a higher flame, you have allowed too much fuel into the priming cup. If the flames are licking at the overhead, they can be extinguished with water.
9. When the preheating flame has died down, open the burner valve a little. If you hear the even sound of escaping gas, then the burner is hot enough, and opening the valve a little more will generally ignite the escaping gas from the priming flame. If there is a sputtering sound, then the burner is not hot enough, and you should wait for it to cool down before repeating steps 7 and 8 above. If the priming flame should go out, and there is the steady sound of escaping gas, then the gas can be ignited and the stove operated.
10. Until the stove has had a chance to heat up, best not to try to use full heat, as the rush of gas may cool the preheater too much.

11. When finished cooking, shut off the stove shut-off valve, then turn off the burner valve gently. Force will injure the burner seat.
12. As an extra safety precaution, you can relieve the pressure in the tank by unscrewing the filler cap and allowing the air to escape. Always re-tighten the filler cap.
13. Avoid allowing alcohol in contact with a hot burner. An explosion may result.

JIFFY REEFING SYSTEM:

The T-37 is equipped for "Jiffy" reefing of the main sail. This reefing system is also called "California", "Slab", "Quick", and a multitude of other names.

Rigging: A line is provided with the vessel for the outboard end of the jiffy reef system. One end of this line should be tied with a bowline to the pad eye on the port side of the boom at its aft end. The line should run from the pad eye up through the first reefing grommet in the mainsail, then down through the cheek block on the adjustable car, and forward to the cleat aft of the winch on the boom.

Operation:

1. Be certain that the main boom topping lift is attached and tied so that the main boom will not fall and hit the deck or crew.
2. Slack the main halyard sufficiently to enable the reefing grommet on the main luff to be slipped over the hook on the spar end of the boom. The halyard should be marked for reference at this point.
3. With the main's grommet engaged in the hook, re-tension the main halyard.
4. Check that the adjustable cheek block on the starboard side of the boom is positioned so that the reefing line will pull the sail out as well as down - somewhere between a 45° and 60° angle to the boom.
5. Release the tension on the outhaul, and cleat off the outhaul pennant.
6. Wrap the reefing line around the winch on the boom, tension the line with the winch, and cleat.
7. Some sailmakers put small grommets in the sail to be used for ties to gather up the excess cloth. If these ties are used, BE CERTAIN that they are all removed before the reef is shaken out.

If additional reefs are needed, the first reef outboard end can be tied off with a sail tie through the grommet on the main and around the boom, and the reefing line re-lead through the next higher grommet.

HOLDING TANK:

Your T-37 is equipped with a holding tank for retention of sewage. The standard system is a "No-Discharge" system, which we believe will be acceptable in all waters.

Sea water is used to flush the head, and the sea water and effluent is pumped into the holding tank by pumping the toilet. A deck fitting is provided through which the holding tank may be evacuated by a shoreside pump out station.

With the standard holding tank, it is not necessary to "pre-charge" the tank by adding water before using the system. Be certain that the pump on the toilet is pumped several additional strokes after waste has been evacuated from the bowl to insure that the effluent is pumped through the hoses and into the tank.

The tank has an inspection port on its top (located under the vee berth port side), which will give indication of the level in the tank. If the toilet should be difficult to pump, check that the holding tank is not overly full. Trying to over fill the holding tank will injure the tank, and may force waste out the air vent fitting.

OPTIONAL RECIRCULATING HOLDING TANK:

An optional arrangement allows the effluent liquid to be used to flush the toilet, which increases the number of uses of the tank. This system must be charged with chemicals before using, initially, and after each pump out. The recirculating system has valves which can be used to pump the toilet with either effluent liquid or sea water.

Typical operation would be to pump sewage out of the toilet using effluent liquid, then change the valves and pump a little sea water into the bowl. Always close the sea water intake valve after use to prevent filling up the tank with sea water.

WINTERIZING:

The holding tank must be pumped out before winter storage, and a small amount of Recreational Vehicle anti-freeze added to the residual water.

III. TARTAN MAINTENANCE SCHEDULE

GENERAL:

Regular preventive maintenance is required to keep any boat in "as new" condition. It starts with the day after delivery and continues throughout the year. The heaviest time commitment is, of course, in the Spring but one should always be observant of the condition of such areas as running rigging, finishes, the engine, head, and other moving parts of gear and tackle. The following comments are intended to serve as an initial guideline. You will no doubt want to develop a check list of your own.

FINISHES:

Fiberglass: Even though fiberglass construction has vastly reduced upkeep, some attention to gel coat surfaces is necessary to maintain the appearance of the finish. After a few years exposure with no protection, the finish may begin to fade or chalk. The annual application of a good commercially available wax (Johnson, Fuller or equal) containing an ultra-violet shield will preserve the appearance of this finish for many years. Generally, an application at the beginning of the season will suffice. Abrasive cleaners (cleanser) should not be used for general cleaning.

Gel Coat: A small quantity of matching gel coat is shipped with each boat. This material should be kept in a cool place, not on board the boat. Generally, the shelf life of gel coat is about 6 months. Even during this period the original gel coat may not perfectly match due to fading. The gel coat is a polyester product, not a paint, and requires mixing a catalyst before use. Patching gel coat is a job requiring some experience, and best results are generally obtained from professional work. A manual on fiberglass repair is shipped with the boat.

Bottom Paint: One coat of bottom paint is applied at the factory. Generally, a second coat applied before launching will afford the best protection. Check with your dealer for a paint compatible to the factory applied paint. A non-compatible paint may lift the factory applied paint.

Bottom paint should be lightly sanded before re-coating. Always wear a respirator when sanding bottom paint - it is toxic. After several coats, it will be necessary to remove the accumulation of bottom paint preferably by sanding with a rotary sponge-backed pad.

Boot Top and Sheer Stripe: Are both painted with polyurethane paint. If touch-up is needed, contact your dealer for the matching paint.

EXTERIOR TEAK:

If left untreated, exterior teak will discolor rapidly, turning a dull grey color. Teak is relatively open grain wood, and eventually, mildew may form in the grain, resulting in a very dark color.

If you wish to maintain the warm brown color, the teak must be kept clean, and oiled. The grain of the teak will raise as the wood is wetted. The job of keeping up the teak will be much easier if the wood is sanded very smooth. Use sandpaper for this purpose, and be careful not to scratch the gel coat. The best routine for bringing back discolored teak is to scrub thoroughly with a teak cleaner and water, allow to dry, sand, then apply multiple coats of a high grade teak oil. Some teak cleaners will stain the gel coat, so be sure to hose off the deck and topsides thoroughly.

A good applicator for teak oil is a small piece of a sponge, perhaps 1" square, by about 3" long, with one end tapered to allow application close to the deck without touching the gel coat. Most teak oils will stain the gel coat, and are difficult to remove, so be careful and clean up drips promptly.

INTERIOR TEAK:

The interior teak has been oiled with Watco Danish Oil Finish, clear. These surfaces should be inspected regularly and touched up if needed.

SAIL, SHEETS AND LINES:

Sails and lines should be removed at the end of each season and stored in a warm, dry place. If it is possible to dry them thoroughly, they should be rinsed with fresh water before storage.

SPARS AND STANDING RIGGING:

The mast should be supported so that it lies straight. A thorough inspection of all wire, swages, splices, pins and fittings should be made prior to winter storage; any necessary replacement or reinforcement should be made before beginning the next sailing season. The boom is anodized and spar painted with a linear polyurethane.

ENGINE AND FUEL SYSTEM:

Check the engine manual for maintenance guidance during the season and for the specific haul out procedures necessary to winterize the engine.

HEAD:

As with the engine, the specific procedures for preparation for winter storage and recommissioning are contained in the Wilcox-Crittenden manual.

BATTERIES:

Remove from boat and store inside. Will need to recharge prior to installing in boat. Do not store batteries on a concrete surface - elevate with pieces of wood. A fully charged battery will freeze at well below zero F; a charged battery freezes at about 20° F.

FRESH WATER SYSTEM:

The water tank and water lines should be complete emptied in preparation for winter storage, and the pressure water pump drained.

WINTER STORAGE:

Cover: If storing outdoors, a winter cover is recommended. It can be as simple as a rectangular piece of canvas forming a tent over the boat. A ridge pole (formed by 2" x 4"'s along the centerline) several feet above the cabin top, well supported at several places along its length, is sufficient to support the center. The stantions can be removed from their sockets, and ropes tied from the ridge pole to the stantion sockets to help support the cover. Use carpeting to pad any areas of chafe. Lash the cover tightly to the cradle, avoiding any metal grommets in contact with the gel coat.

Bilges: Be sure to pump the bilge completely dry.

Ventilation: Leave the dorade vents in place and open so the boat can get air during the winter.

Cradle: Make sure that the boat is adequately supported and that any suspected weakness has been reinforced. The keel of the boat must rest solidly on the main beam. The vertical risers are not intended to carry the load, merely to stabilize the boat.

RIGGING LIST

STANDING RIGGING:

<u>Qty. Description</u>	<u>Wire Size</u>	<u>End Fitting-Top</u>	<u>End Fitting-Bottom</u>
1 Headstay	5/16"	Marine Eye	Marine Eye
1 Backstay	5/16"	Marine Eye	#585 5/8" Turnbuckle
2 Upper Shroud	5/16"	Marine Eye	#585 5/8" Turnbuckle
2 Fwd. Lower Shr.	1/4"	Marine Eye	#585 1/2" Turnbuckle
2 Aft Lower Shr.	1/4"	Marine Eye	#585 1/2" Turnbuckle

RUNNING RIGGING:

<u>Qty. Description</u>	<u>Material</u>	<u>Size</u>	<u>End Fitting</u>
1 Main Sheet	Br. Dacron	7/16"	Eye splice - Burn
2 Genoa Sheets	Br. Dacron	9/16"	Burn - Burn
1 Outboard and Quick Reef	Br. Dacron	3/8"	Burn - Burn
1 Main Halyard	Prestretched Dacron	3/8"	Splice in #390 Shackle - Burn
1 Genoa Halyard	7 x 19 SS Wire	3/16"	Thimble & micropress in #721-3 Snap Shackle
	Br. Dacron	7/16"	Burn
1 Main Boom Topping Life	Vinyl coated 7 x 19 Wire	1/8"	Thimble micropressed in each end w/FG-365 shackle one end
1 Main Boom Topping Pennant	Br. Dacron	5/16"	Eye splice around FG-365 shackle - Burn
1 Masthead Flag Halyard	Nylon	1/8"	Burn - Burn
1 Centerboard Pennant	Pre-stretched Dacron	3/8"	Burn - Burn
1 Centerboard Tag Line	Br. Dacron	3/8"	Splice around deck mtd. eye - Burn

MASTER CB ①
30A

GROUND
BLACK -
EXCEPT AS
NOTED

CABIN L.T.S. ②
10A
#14GY 2COND (RED+) PORT
#14GY 2COND (RED+) STB
#14GY STERN L.T.
#14GY BOW L.T.
#14 RED
#14 PURPLE - COMPASS &
NAV. INST. L.T.S.
WATER PRESS. #12 DK. BROWN
INHIBIT AUTO BILGE
SWITCH
#12 LT. BROWN
SUMP PUMP
#12 YELLOW
BLOWER
#12 YELLOW
ALL GROUNDS ⑦

OA
BREAKERS
ENG. INST.
PAN
#14-O
ANCH.
#14GY
MAST
FD
L.T.
#12 W
GND

NOTE: ALL GROUNDS
RUN INDIVIDUALLY

MAST WIRING

MAST L.T. OR
STEAMING L.T.

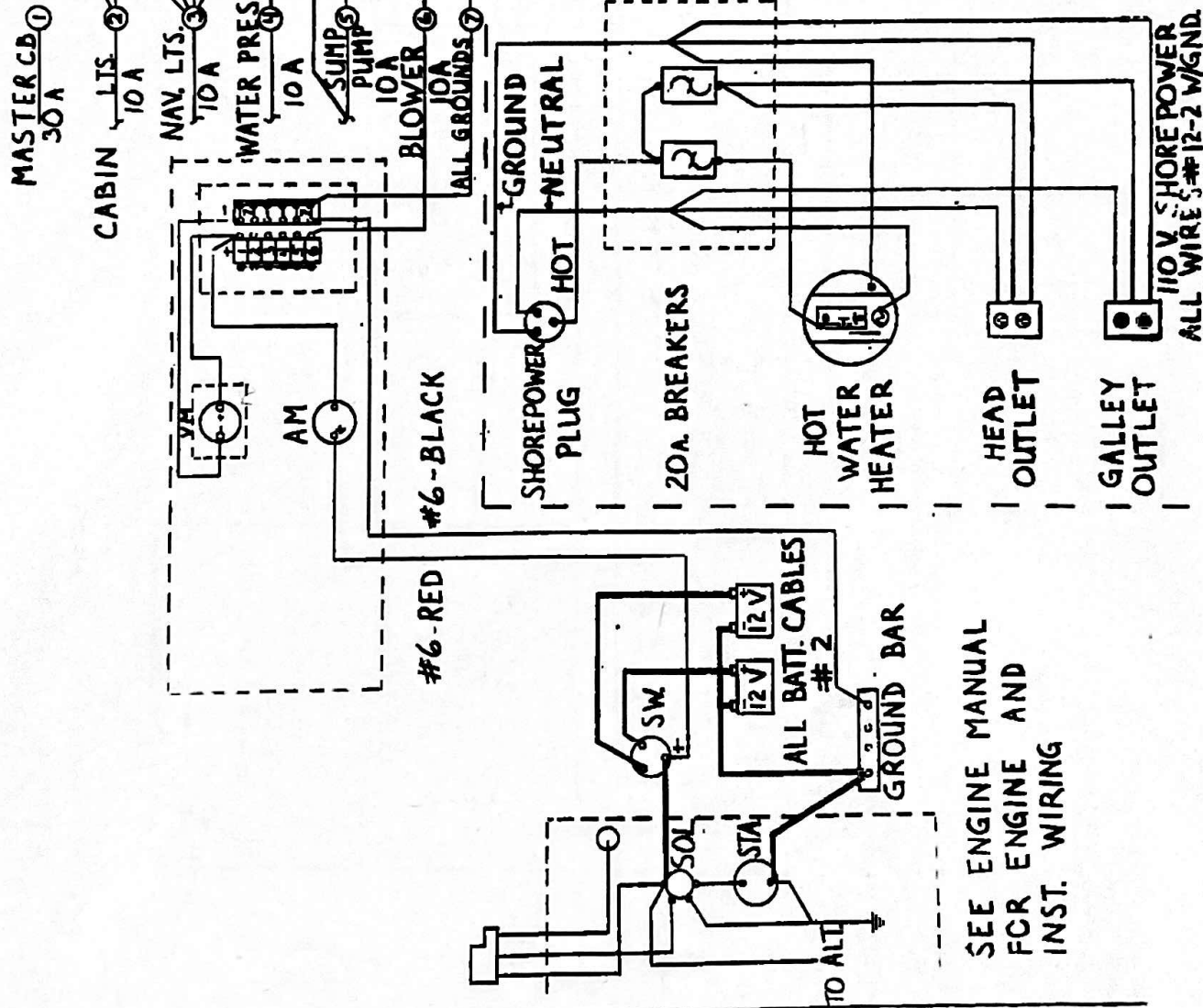
FD & ANCH. L.T.S.
OPT.

NOTE: SHOREPOWER IS TOTALLY
INDEPENDENT FROM SHIP'S
GROUND. OBSERVE POLARITY
WHEN HOOKING UP SHORE
CORD.

TARTAN 37 WIRING

A.F.C.

OCT. 76

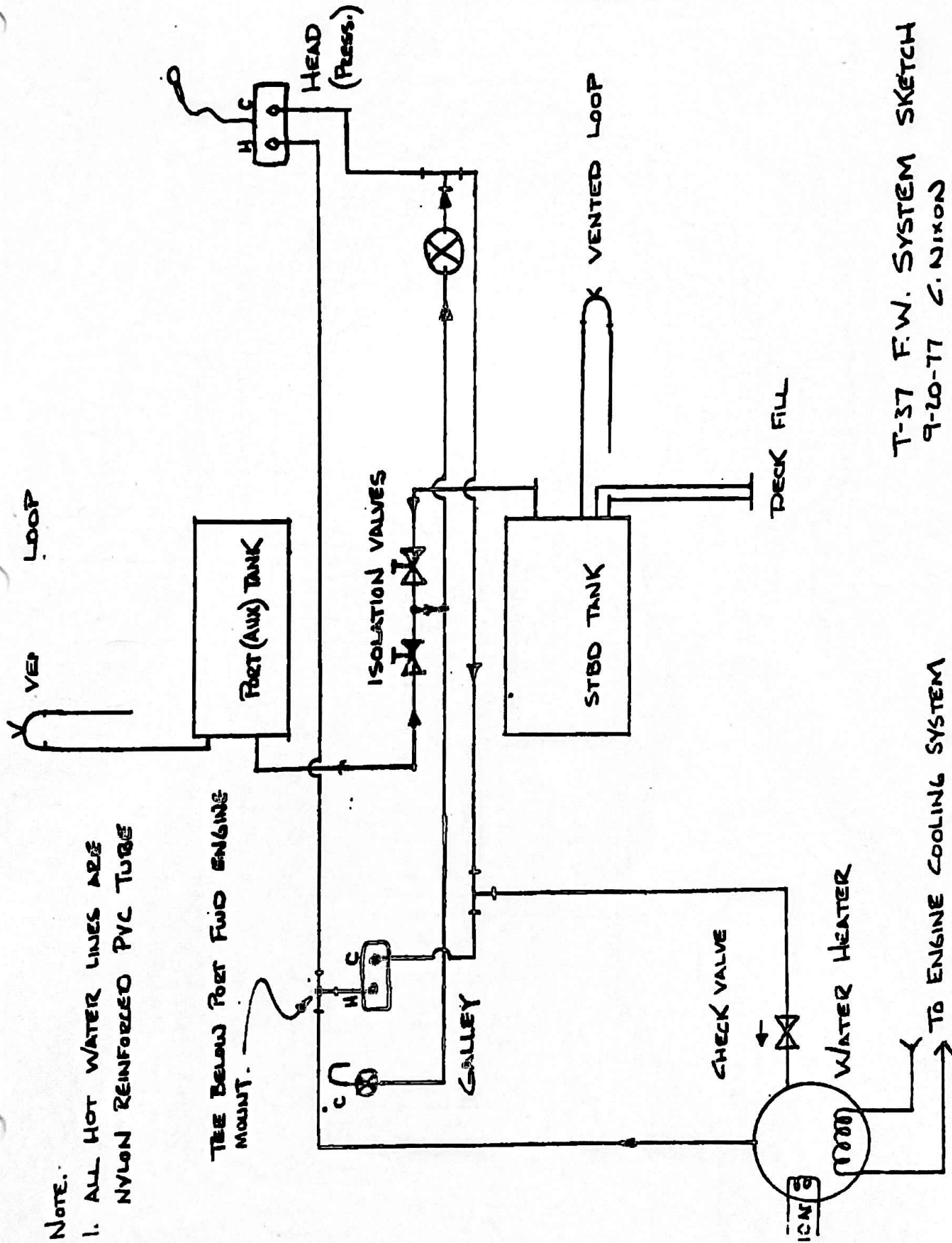


SEE ENGINE MANUAL
FOR ENGINE AND
INST. WIRING

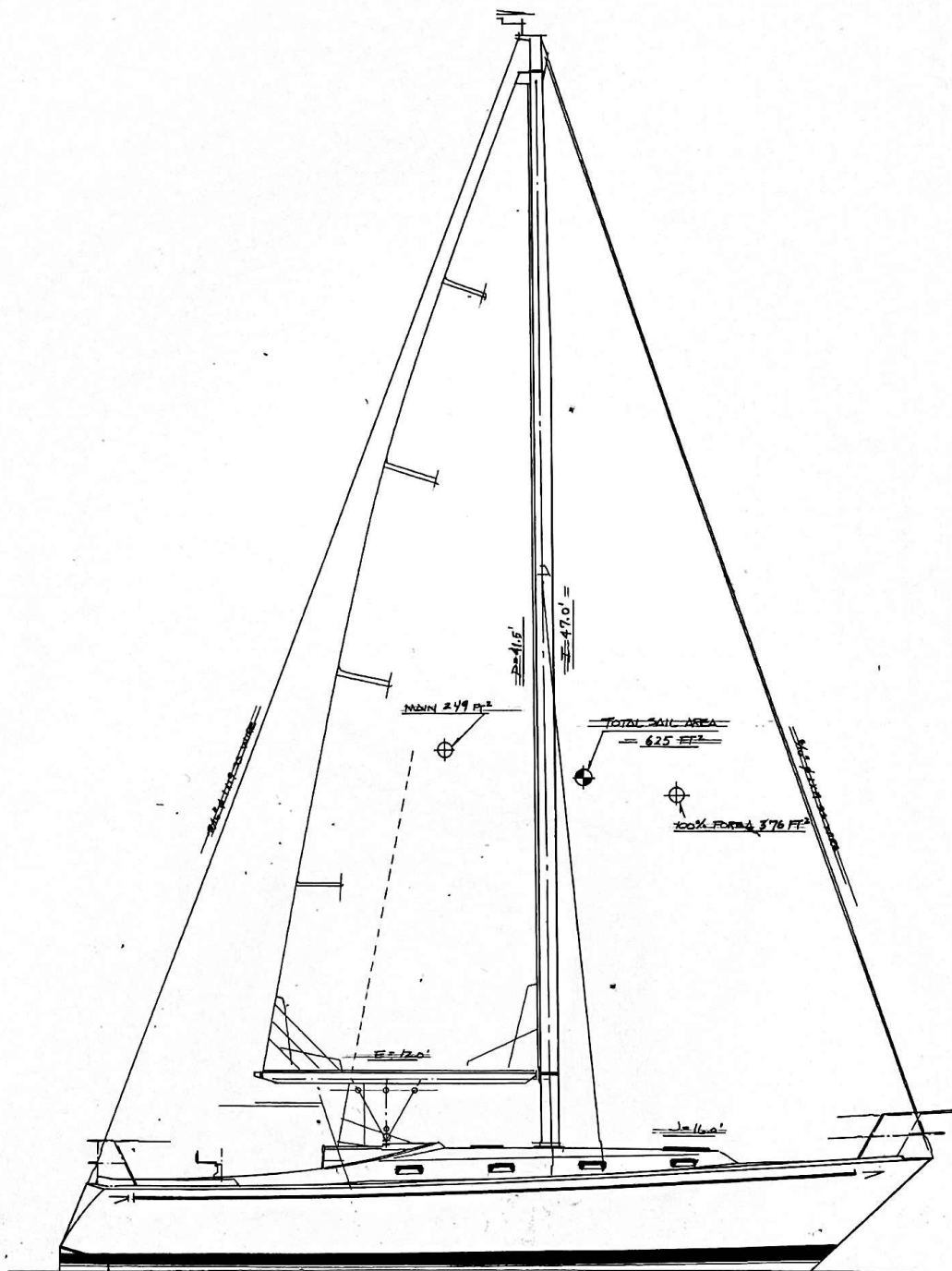
110V SHOREPOWER
ALL WIRE #12-2 W/GND

NOTE:

1. ALL HOT WATER LINES ARE NYLON REINFORCED PVC TUBE



T-37 F.W. SYSTEM SKETCH
 9-20-77 C. NIXON
 REV A - HULL # 107 d up 4-12-78



TARTAN 37
SILVER CENTERBOARD STD RIG
24' ODDER 25' 3/4" LOA 7'0"
TARTAN MARINE CO.
210 RIVER ST. GRAND RIVER, OHIO